Problem Decompositions

The Problem A

The Ministry of Health wants to make data-driven decisions, but lacks a unifying layer that makes the data visible, comparable, and actionable, undermining effective and timely decision-making.

Who you would talk to in Week 1 and what you would ask
My first approach will be to conduct a diagnosis to understand what makes the data messy. Given that the Ministry of Health is in charge of this layer, here are the questions I would ask the director:
Key personnel: MoH Director
- How does data get into the DHIS2 for export?
- Are there facilities that never send in their report?
- On average, how long does it take to receive reports from the facilities?
- Which of the 3 (clinic, hospital, rural facility) takes the longest to send their data?
- Which of the 3 (clinic, hospital, rural facility) do you spend more time refining and cleaning?
- What corrections happen after data collection?
- What part of the compilation process takes the longest time? What part is manual?

Direct Observation
- I would ask the MoH team to walk me through the process of DHIS2 export to Excel, and the pre-processing measures they take, including data cleaning, augmentation, normalisation, and addition.
- I would watch their bulletin generation process: are bulletins created from scratch every time, or are there templates? What part can be automated now, and what process or workflow needs an overhaul or a new system? These questions would be answered by observation.
- I would understand the computations that go into calculating top facilities, maternal health indicators, and trend analysis.
- I would like to walk backwards from the last bulletin to confirm how data is exported from DHIS2 and prepared to make up the bulletin. This includes the conversations had, the bottlenecks in the process, and the tools currently used. This helps me know first-hand what the workflow is and where the gaps lie.
- I would like to understand what information makes it into the bulletin.

Assumptions about the MoH data mess
- Inconsistent submission may be due to power, spotty internet, or the manual requirement of handling paper-based reports.
- Data is a mess because some facilities may be inconsistent with their submissions.

Patterns I See:
- Data fragmentation: Data is scattered across multiple sources (DHIS2, CommCare, OpenMRS, and the Health Tracker EMR).
- Data is incomplete/unorganised: given that up to 175 facilities are not fully digitised, data will remain uncaptured in the bulletin and dashboards, worsened by unreliable power. Some facilities may not be submitting their reports.
- Data is inconsistent: given that data may come from a buggy EMR, it may be duplicated, contain inaccurate entries (e.g., 'Male is pregnant'), use an inconsistent schema (across multiple versions of the EMR, the shape of the data schema can change), or have missing values (incorrect, duplicated, or inaccurate formats generated by a buggy EMR).
- Data is slow to access: data is stale, given that some facilities are without power or may experience spotty internet, which will delay data transfer. For paper-based facilities, data would remain captured but untransmitted until physically delivered, which takes time and may be delayed.
- Data compilation is manual and not automated, so data cleaning and preparation may be taking the bulk of the MoH's time.

The Ministry needs a full-coverage health information system that aggregates data from these different sources for processing and analysis, including report generation on the top-performing facilities by month with comparison against the previous month.

The Problem B

A District Health Officer sits between the facilities and the national MoH. They act as the operational manager for all health facilities in one district.

Their core responsibilities may include:
Resource allocation: where do drugs, staff, supplies go this week
Supervision: are facilities operational, staffed, functioning
Outbreak response: spot a disease cluster early, mobilize a response
Stock management: catch stockouts before a facility runs dry
Reporting upward: roll district numbers up to the national level

The District Health Officer (DHO) cannot manage the district effectively if the information is stale. Learning about operational and time-sensitive issues like a stockout or outbreak after 3 weeks can directly cause harm to patients or increase a community's risk of exposure to an outbreak.

Who you would talk to in Week 1 and what you would ask
There is no real-time system for report/data dissemination from facility to the district officer.
Questions to ask the DHO:
- How do Clinics and Hospitals send data to DHIS2? Is it automated, or does it require manual preparation?
- Do you find that you experience delay from Clinics and Hospitals due to connectivity and power supply?
- How do you digitise the rural facility data? Do you have to conduct manual data entry?
- How are data delivered from rural facilities? Are they physical?
- What is the average time it takes to receive data from rural and digitised facilities?

- At what point between the facility and DHIS2 is most of the reporting delay introduced?
- Which of the 3 types of facilities do you find have the most challenge with data submission?
- Are there facilities that deliver reports physically?

Assumptions
- For rural reporting that requires physical delivery, data collation might be handled on a given day of the month. Because of the physical requirement, it makes sense for a facility to collate the data and send it off in one day.

Patterns I See:
- Power and connectivity: For digitised hospitals and clinics, they may still face some level of delay in reporting to the DHOs, given that connection and power are required to transmit data.
- Collection problems: Facilities that are managed on paper may require physical means to report status: either the DHO visits each facility, or the facility manager physically reports to the DHO. Given that there are up to 175 facilities, the physical constraint may be a large contributor to slow or stale data.
- Integration problems: If integration pipelines for sharing data from clinics and hospitals are not fully established, data transfer could be delayed due to the lack of a clear path for data dissemination to the DHO.
- Manual data entry: for paper-based facilities, the DHO team may be required to conduct manual entry of the data, which takes time and requires human hours.

The Problem C
The HIV and TB programmes are managed on CommCare, but we can't identify patients with HIV/TB co-infections.

Who you would talk to in Week 1 and what you would ask
- How do the HIV and TB programmes identify the same patient?
- Are HIV and TB recorded in the same CommCare application or in separate applications/projects?
- If the data is already collected and exists in independent programmes, what prevents linking it today?
- Walk me through how a patient who has both HIV and TB is represented across the two programmes today?

Patterns I See:
- The HIV and TB programmes may be deployed on separate CommCare applications, with two different forms and databases.
- The same patient with different identification numbers across programmes.

Assumptions about the SandTech product suite
- HOS Data Model: a transformation engine that maps incoming data to a canonical model and applies validation logic before storing it or passing it on for analytics and AI. I also assume it has an ingestion layer that lets it process data from multiple sources.
- Health Atlas: supports location-based decision-making and can be used for disease surveillance, such as tuberculosis and HIV intervention, as well as immunization surveillance.
- HealthInsight Engine (AI analytics): sits on top of the HOS Data Model and handles forecasting, prediction, and summarising findings for the Ministry.
- Outcome Tracker: likely provides outcome analytics and probably supports reporting or exporting them. I also assume it provides the metrics used in report generation, such as quarterly bulletins.
- Apache Superset (analytics templates toolkit): sits on top of the data from the HOS Data Model and provides templates to visualise analytics and generate bulletins and reports. Example templates include a monthly bulletin layout, a maternal health dashboard, a quarterly PDF report, and a facility performance dashboard.

Problem Selection

I would choose Problem A because it has the clearest structure and the most available information: known data sources, an existing workflow to observe, and a standard output. Given that the Ministry already produces the bulletin manually, the team would not be building from scratch but would be identifying gaps to close and workflows to automate.

Concrete Reasons:
- Major pain point directly affecting the MoH decision-making.
- The data source is known: DHIS2 Excel exports.
- The workflow is directly observable and can be automated.
- The metrics for the Bulletin are clearly defined.

The sprint would focus on exposing data quality issues, exposing workflow gaps, generating usable output templates, and turning that manual process into a repeatable automation.

This is the quickest way to gain trust from the MoH and gain momentum in results. The lessons learned can also be reusable information for later work within the district office and facilities.

Compared to Problems B and C:
- Both problems may require direct facility observation, which takes time to access physically.
- In Problem B, the workflow can vary across facilities. Creating value across multiple facilities might require new tooling, or close observation and incremental solution deployment. This takes time.
- In Problem B, "Real-time" facility status depends on connectivity, power, reporting workflows, and source system reliability. Rural facilities are paper-only, which limits real-time visibility.
- Problem B would require a new system for digitising rural facilities, which is usually accompanied by training of hospital staff. This might take more than 6 weeks to deploy and train.
- Problem C requires patient matching, privacy controls, facility ownership, and cross-programme coordination. I would defer it because it requires solutions to be deployed at the facility level; given the number of facilities, it would also require more time.

Specific Measurable Outcome
By the end of 6 weeks:
- Ingests the current DHIS2 export format.
- Calculates at least the core bulletin metrics: top facilities by volume, maternal health indicators, facility performance, and quarter-over-quarter trends.
- Produces a usable bulletin output, such as an HTML report.
- Reduces manual compilation effort and cuts down hours in bulletin preparation.

Out Of Scope For The First Sprint
- Replacing DHIS2, HealthTrack, OpenMRS, CommCare, or any existing source system.
- Digitising the 175 paper-only rural facilities.
- Building real-time facility monitoring for all districts.

Week 2 Validation List
- Understand the existing workflow, where the 40 hours are spent, and identify the biggest bottlenecks.
- Verify the quality and completeness of the DHIS2 exports and identify any manual steps the MoH takes to clean the data.
- Review the existing bulletin templates, calculation rules, for previous metric.
- Verify that the Analytics Template Toolkit can generate a bulletin that matches the Ministry's current format.
- Confirm that the Health Outcome Tracker calculates the required metrics and produces the expected results as compared to the previous bulletin.
- Validate that the HealthOS Data Models correctly transform and standardise the DHIS2 data for downstream reporting.

Rapid Prototyping

Which Sand products you will use and why

- HealthOS Data Models: To transform and standardise the raw DHIS2 exports into a consistent healthcare data model that downstream analytics can consume.
- Health Outcome Tracker: Compute and manage healthcare indicators like delivery volume, mortality rates, and trend analysis.
- Analytics Template Toolkit: To generate a reusable Quarterly Health Bulletin template and present the computed indicators as an HTML dashboard or PDF report. This would reduce any bulletin preparation time.

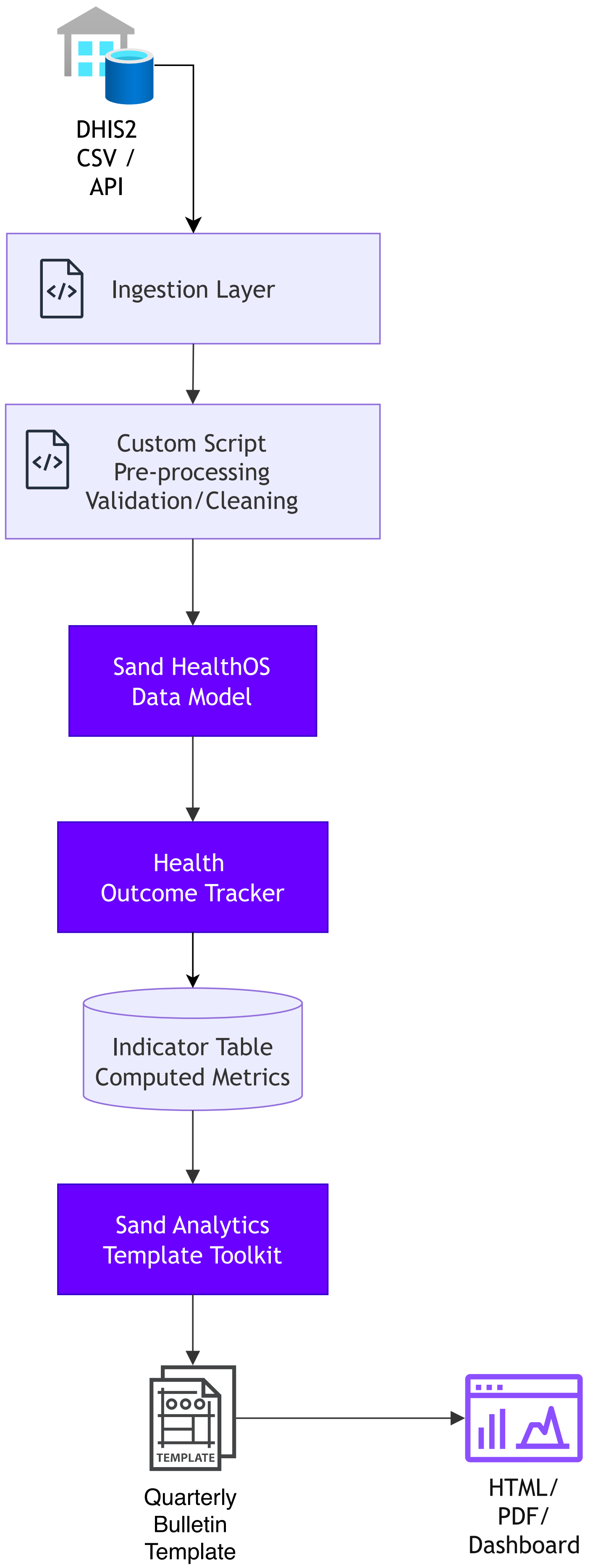
What you would show in Week 3 (what works, what is broken)

What works:

- Ingests the DHIS2 export.
- Calculates the bulletin metrics: top facilities by delivery volume, neonatal mortality rate, facility performance (readiness + reporting completeness), and quarter-over-quarter trend.
- Generates a simple HTML bulletin (or dashboard) displaying the computed metrics.

Known limitations / What's broken:

- In this implementation, we have not made use of Sand Tech solutions, so the data standardisation layer will be implemented with them later.
- ANC visit cannot be calculated because the supplied dataset does not include ANC visit data.
- Delivery is used as a proxy value for patient volume.
- Reporting timeliness cannot be calculated because submission timestamps or expected reporting dates are not available.
- Trend analysis is only implemented for the clinical neonatal dataset, since it is the only dataset containing monthly observations.
- The bulletin uses a simple template intended to demonstrate the concept and not a production-ready design.


DHIS2
CSV /
API
Ingestion Layer
Custom Script
Pre-processing
Validation/Cleaning
Sand HealthOS
Data Model
Health
Outcome Tracker
Indicator Table
Computed Metrics
Sand Analytics
Template Toolkit
TEMPLATE
Quarterly
Bulletin
Template
HTML/
PDF/
Dashboard

Neonatal Health Bulletin

A small pipeline that turns the DHIS2-style CSV exports into a quarterly bulletin.
It reads the raw data, computes a few facility metrics with Danfo.js, and renders them as charts in a plain HTML page.

Project

https://github.com/kerry-okpere/SandTech

Setup

You'll need Node (v18+). Then:
For setting up environment:

```
npm install
```

For running the pipeline:

```
node index.js
```

run this to view in the browser:

```
npm run serve
```

Go to http://localhost:8000/bulletin.html.

Updating the data

To update the data, just swap the files and regenerate:
1. Replace `data_pipeline/data/clinical_neonatal.csv` with the new export (keep the same filename and column headers).
2. Run `node index.js` again, to override `bulletin.json` with the new numbers.
3. Refresh the browser (or `npm run serve` again).

Same goes for any of the other CSVs in `data_pipeline/data/`.

Layout

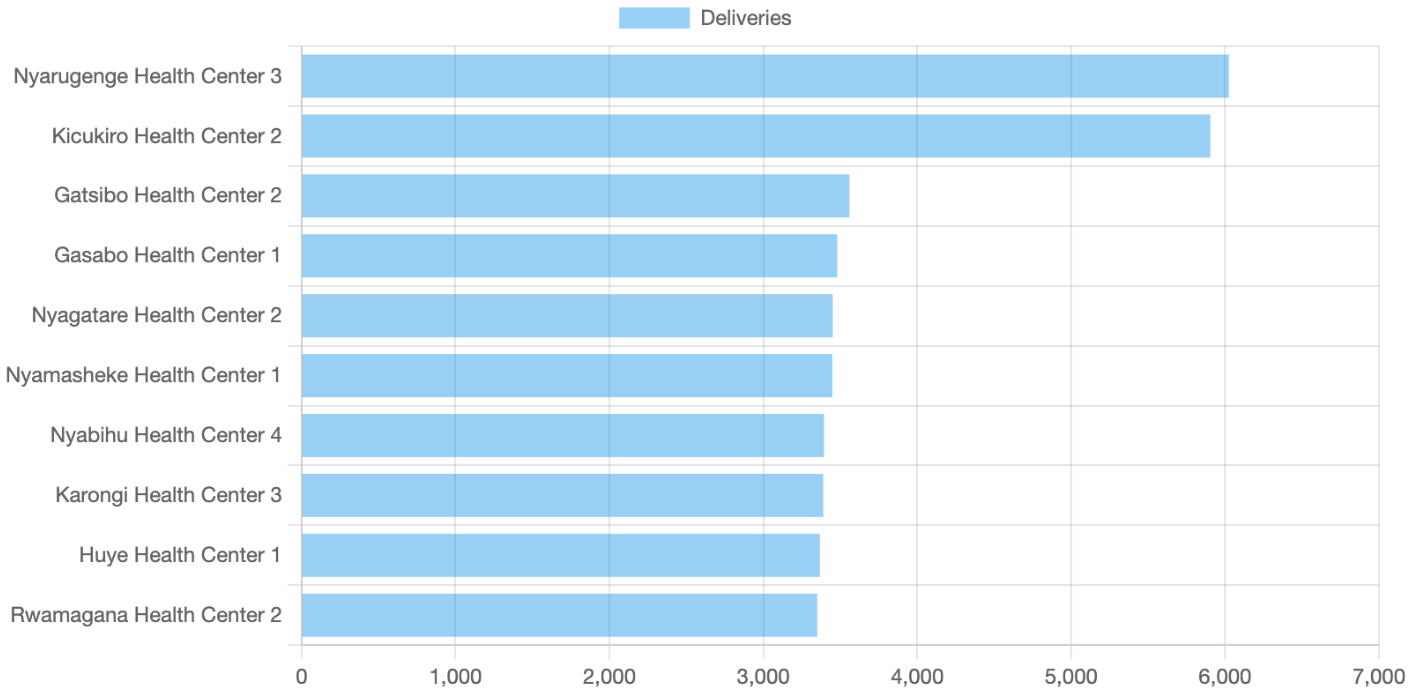
```
data_pipeline/
  data/            the DHIS2 CSV exports
  metrics/
    index.js       DHIS2DataLoader + the four metric functions
index.js           runs the metrics and writes bulletin.json
bulletin.html      the dashboard (fetches bulletin.json)
```

Project

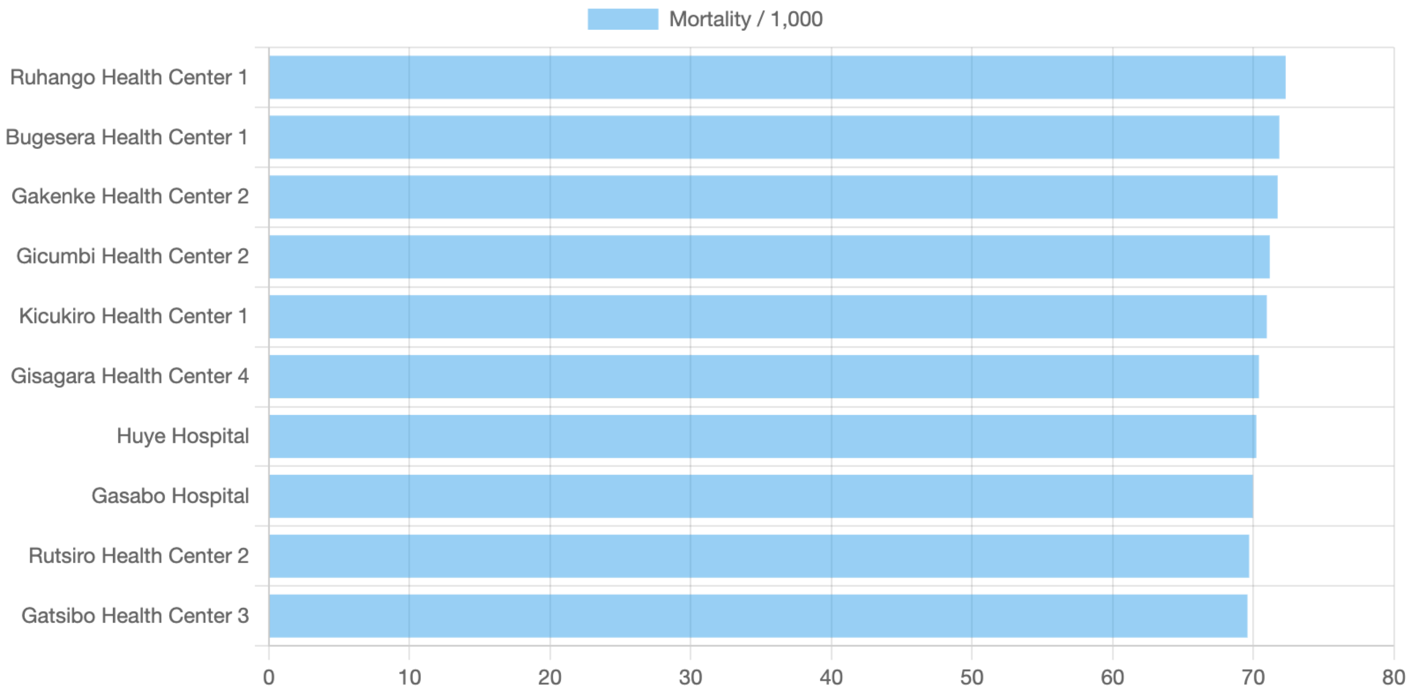
https://github.com/kerry-okpere/SandTech

Neonatal Bulletin 2024

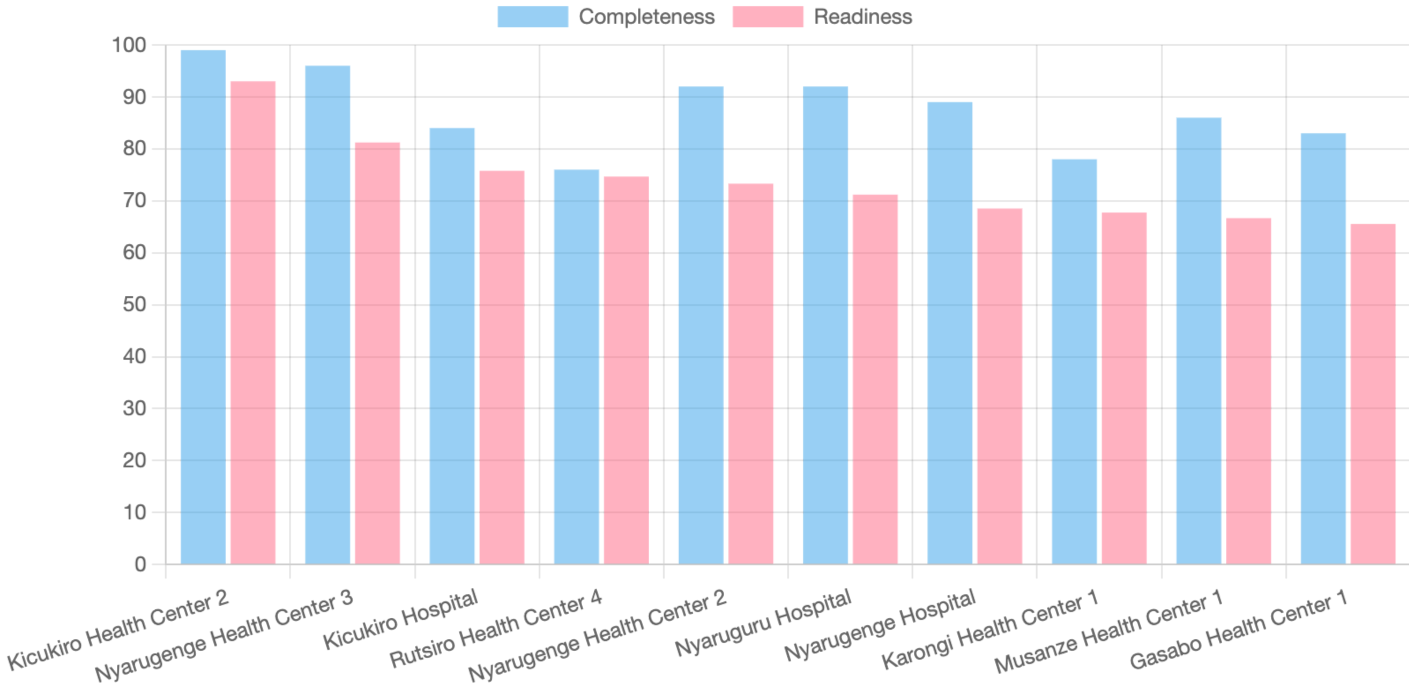
Top 10 facilities by deliveries



Highest neonatal mortality



Reporting completeness vs readiness



Quarterly mortality trend